

## CERN70 Community Event: your questions answered

Can I still register? Which roads and car parks will close when? How do I get my food vouchers? We answer these questions and more ahead of the 17 September event



An [interactive village map](https://maps.cern.ch/cern70), accessible via the [CERN Campus App](#), shows facilities and services available throughout the 17 September CERN70 Community event, including parking areas and shuttle drop-off points.

On Tuesday, 17 September, we will enjoy music, food, birthday cake and more to celebrate and thank our unique CERN community and our families.

To prepare for the [CERN70 Community Event](#), we answer some key questions:

### 1) REGISTRATION: Can I still register to attend?

Yes, by popular demand the registration deadline has been extended to Monday, 9 September at 11.59 p.m. CEST. Click [here](#) to register.

### 2) FOOD VOUCHERS AND ACCESS CARDS: How do I get my food vouchers and CERN70 Guest access cards in advance?

Bracelets with [food and drink](#) vouchers and CERN70 Guest access cards will be distributed to registered persons at the CERN Community Support Centre ([Building 33](#)) from 9 to 17 September, Monday to Friday, from 8 a.m. to 12 noon and from 1 to 5 p.m.

You only need a CERN70 Guest access card if you are bringing a guest who does not have a CERN access card. Visitor passes will not be allowed.

If you and your guests all have CERN access cards, you can collect bracelets with food and drinks vouchers either at Building 33 from 9 September or at additional locations from 2 September. Click [here](#) for full details.

## A Word from Christopher Hartley Streamlined procurement now helps us all

Changes to CERN's Procurement Rules were long overdue. Here's why and how they'll help our CERN community and our Member and Associate Member States

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### 3) SITE INFORMATION: Which roads and car parks are closed when?



*The area shown in yellow will be impacted in the days before and after the CERN70 Community Event.*

The image above shows, in yellow, the area that is impacted by the CERN70 Community Event.

#### Friday, 6 September

- **Route Bohr** closed for the day

#### From Tuesday, 10 to Saturday, 21 September

- **Route Bohr** closed
- **Car parks** [Les Cèdres](#) (next to the Staff Association) and [Greinacher](#) (under the power lines) **closed**
- **Car park** [Les Cerisiers](#) (next to Buildings 4 and 5) **open** but only accessible from Entrance A
- **Curie-Bohr roundabout** partially blocked, with traffic lights to access the car park of Building 40
- **Alternative car parks** close to shuttle stops are visible on this [interactive map](#)

#### Monday, 16 September

- **Car park Building 40** closed from 7 p.m.

#### Tuesday, 17 September

- **Car park Building 40** closed all day
- **Car park Les Cerisiers:** Vehicles cannot exit this car park between 5 and 11 p.m. Entrance A will reopen at 11 p.m. for vehicles that need to exit
- **Alternative car parks** close to shuttle stops are visible on this [interactive map](#)

### 4) ACCESS AND MOBILITY: How can I get to the event?

**Pedestrians** can enter the Meyrin site through all CERN entrances with their CERN access card or a CERN70 Guest access card. Note that visitor passes will not be allowed.

**Cyclists** can enter the Meyrin site via Entrances B, C and E. There will be extra bicycle parking available at the Mobility centre outside Entrance A.

**Shuttle buses** will operate from across the Meyrin site and from the Prévessin site. This [interactive map](#) shows car parks located close to shuttle stops.

**Vehicles** cannot enter or exit Entrance A from 5 to 11 p.m. Entrances B and C will be open for vehicles in both directions throughout the event. Entrance E will be open for vehicles in both directions from 5 to 11 p.m.

More details on the [CERN Community Event website](#).

### 5) ACCESSIBILITY: What accessibility options are available?

There will be a platform for wheelchair users to be able to see the stage, dedicated parking, shuttles and seating available. If you have accessibility issues that were not noted when you registered, please contact [cern.70@cern.ch](mailto:cern.70@cern.ch).

### 6) ENTERTAINMENT: What activities can I expect?

Visit the [CERN Community Event website](#) for full details of stage performances, activities for children and more.

### 7) DURING THE EVENT: What if I have a question or problem during the event?

You can visit an information point at the event or check the information banners on the [CERN Campus App](#). In addition, the CERN Service Portal will be open for queries via 00 41 22 76 77777. For emergencies, contact [00 41 22 76 74444](tel:0041227674444).

We look forward to coming together to celebrate CERN's unique community, rich history and bright future on 17 September.

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## A Word from Christopher Hartley

### Streamlined procurement now helps us all

Changes to CERN's Procurement Rules were long overdue. Here's why and how they'll help our CERN community and our Member and Associate Member States

The CERN Council unanimously approved changes to the Procurement Rules in June. This is a major success that affects not only the CERN community but also industry in CERN's Member and Associate Member States. But why were the changes needed and how will they help us all?

#### Long-overdue changes

Prices in Switzerland have increased fivefold since 1965, yet that was the last time procurement thresholds were set by CERN's Finance Committee. As a result, contracts of relatively low value required complex tendering procedures. This impacted all stakeholders, even discouraging bids in periods of high demand for supplies or services.

Moreover, by setting a [main objective for 2021–2025](#) of increasing return on investment to its Member and Associate Member States, CERN reinforced its commitment to improving the balance on industrial contracts. Work began in 2022, benchmarking CERN's Procurement Rules against EIROforum organisations and the European Commission. Excellent collaboration between Procurement, Finance, the Legal Service and – crucially – the CERN Industrial Liaison Officer (ILO) Forum led to the successful Council presentation in June where the proposed changes to the Rules were unanimously approved.

#### Thresholds doubled

Benchmarking showed that CERN's thresholds were much lower than in any other organisation. As a result, all but one threshold are now doubled (see table). This optimises resources, reduces paperwork and increases flexibility across CERN and for our suppliers.

|                                                           | Previous threshold in CHF | New threshold in CHF |
|-----------------------------------------------------------|---------------------------|----------------------|
| CERN community can make their own orders                  | < 1000                    | < 2000               |
| Price enquiry (DO) handled by Procurement service as from | 5000                      | 10 000               |
| Three bids required as from                               | 5000                      | 10 000               |
| Single source subject to Finance Committee approval       | 200 000                   | 200 000 no change    |
| Market survey/invitation to tender required as from       | 200 000                   | 400 000              |
| Competitive tender subject to Finance Committee approval  | 750 000                   | 1 500 000            |

#### Widened use of limited tendering

Previously, limited tendering was possible only if a Member State had an industrial return coefficient below 0.4. Now, CERN may, if there is sufficient competition, contact only firms proposing supplies and services originating in the 12 Member States with the lowest industrial return coefficient (list reviewed annually). This provides greater flexibility to improve return on investment.

#### Best value for money

The Finance Committee no longer needs to approve in advance whether a contract will be awarded on a lowest-compliant basis or on a best-value-for-money (BVFM) basis. This saves time, encourages competition and enables environmental aspects to be part of the adjudication criteria. This important change reflects CERN's approach to environmentally

responsible procurement, while retaining a key focus on pricing and other requirements. All these changes mean more efficient, more attractive procurement. Yet what hasn't changed is our basic principles. Contracts must meet technical, financial, environmental and delivery requirements, keeping overall costs for CERN as low as possible while striving for balanced industrial return for all the Member and Associate Member States.

Our upcoming [CERN Industry Webinar](#) on 18 September will outline how to do business with CERN, as well as our tendering process and procurement needs. Spread the word to your industrial contacts and encourage them to [register](#).

*Christopher Hartley,  
leader of the Industry, Procurement and  
Knowledge Transfer Department*

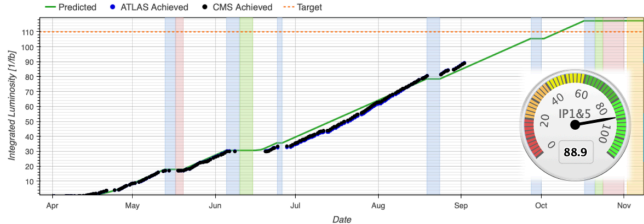
# Accelerator Report: LHC Run 3 achieves record-breaking integrated luminosity

On Sunday, 1 September, the LHC integrated luminosity for Run 3 (2022–2025) equalled that of Run 2 (2015–2018)

LHC Run 3 began in 2022, with the recommissioning of the LHC at 6.8 TeV, resulting in the delivery of 39.7 fb<sup>-1</sup> of integrated luminosity, surpassing initial expectations. Luminosity is an important indicator of the performance of a collider: it is proportional to the number of collisions that occur in a given amount of time. The higher the luminosity, the more data the experiments can gather to allow them to observe rare processes.

In the second year of Run 3 (2023), we faced challenges due to faults that required [extended repair and recovery periods](#), resulting in the delivery of only 31.8 fb<sup>-1</sup>.

This year, though, luminosity production has exceeded expectations, reaching, on Monday 2 September, 88.9 fb<sup>-1</sup> for ATLAS and CMS, 7.6 fb<sup>-1</sup> for LHCb and 45.6 fb<sup>-1</sup> for ALICE. With approximately six weeks of luminosity production still ahead of us, the 2024 target of 110 fb<sup>-1</sup> for ATLAS and CMS is well within reach.



The planned and achieved integrated luminosity for ATLAS and CMS, which reached 88.9 fb<sup>-1</sup> on 2 September. (Blue indicates the machine development time, green the technical stops, red the lead-ion run and yellow the YETS 2024–2025. The red dotted line indicates the 2024 target of 110 fb<sup>-1</sup>).

During long shutdowns (LS) and year-end technical stops (YETS), performance enhancements are implemented and commissioned with beam. Additionally, throughout the year, the LHC teams continuously explore performance margins, ultimately leading to increased luminosity production. Naturally, the total integrated luminosity each year also depends on the duration of the yearly proton run.

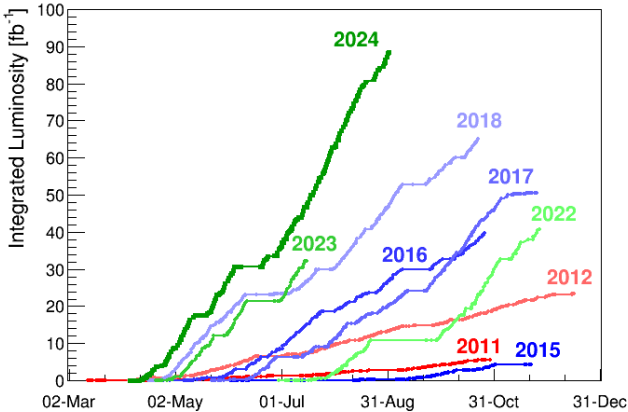
In 2022, there were effectively 70.5 days of proton physics, yielding an average integrated luminosity production of 0.56 fb<sup>-1</sup> per day. Despite the setbacks in 2023, where faults necessitated extended repair and recovery periods, the average integrated luminosity production reached 0.71 fb<sup>-1</sup> per day over 47.5 days of proton physics. As of Monday, 2 September, a total of 88.9 fb<sup>-1</sup> has been achieved in 107 days of proton physics, resulting in an average integrated luminosity of 0.83 fb<sup>-1</sup> per day. And so, the integrated luminosity for Run 3 has already equalled the total integrated luminosity of the entire four-year duration of Run 2, as can be seen in the table below.

| Run          | Years              | Average integrated luminosity ATLAS/CMS [fb <sup>-1</sup> ] |
|--------------|--------------------|-------------------------------------------------------------|
| Run 1 total: | 2010 – 2011 – 2012 | 29.2                                                        |



|                          |                                |              |
|--------------------------|--------------------------------|--------------|
| Run 2 total:             | 2015 – 2016 –<br>2017 – 2018   | 159.8        |
| Run 3 total<br>(so far): | 2022 – 2023 –<br>2024 (– 2025) | 160.4        |
| <b>Total:</b>            |                                | <b>349.4</b> |

Overview of the integrated luminosity per run as of 2 September, clearly showing that the integrated luminosity of Run 3 (160.4 fb<sup>-1</sup>) has already surpassed the Run 2 integrated luminosity (159.8 fb<sup>-1</sup>).



Overview of integrated luminosity as a function of the date for each year of LHC running, with 2024 greatly exceeding all other years.

The LHC schedule has another 40 days of proton physics planned before transitioning to lead-ion physics. Theoretically, this could allow for the accumulation of an additional 33 fb<sup>-1</sup>, potentially exceeding the 2024 target by 12 fb<sup>-1</sup>. However, achieving this will depend on both the availability of the LHC and its subsystems and on the stable beam ratio that can be maintained in the coming weeks.

Rende Steerenberg

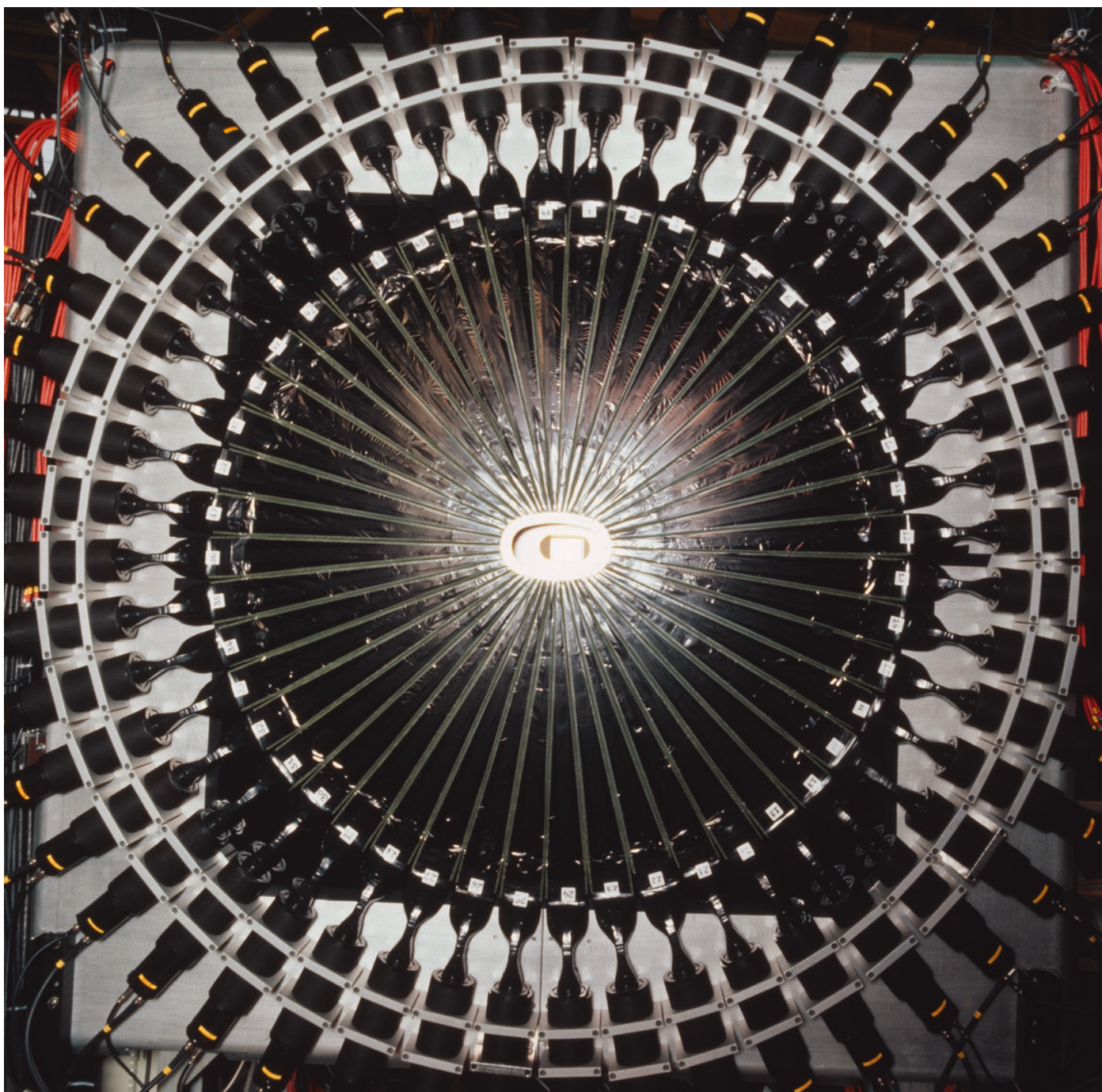
## CERN70: Into the antiworld

Part 16 of the [CERN70](#) feature series. Walter Oelert led the team of researchers who produced the world’s first atoms of antihydrogen in 1995

For each particle, there exists an antiparticle with opposite properties, in particular electric charge. This has been well established, ever since Paul Dirac's theoretical predictions in the late 1920s. Over the three decades that followed, scientists discovered the constituents that would make up an antimatter atom: the antielectron (or positron), the antiproton and the antineutron. But it was not until 1995 that a CERN collaboration succeeded in making, for the first time in the world, a few atoms of antihydrogen, each consisting of one positron and one antiproton.

These first antihydrogen atoms were observed by the PS210 experiment, in the [Low Energy Antiproton Ring \(LEAR\)](#). This experiment was a real challenge, both for the experimenters and the LEAR operation team, but they succeeded, although the antimatter atoms were produced with nearly the speed of light and could not be used for further investigations of the properties of antimatter. On 4 January 1996, CERN and the institutes participating in the experiment issued a [press release](#) announcing the observations. The news, which seemed to come straight out of a science fiction film, went round the world.

This success, and the considerable interest shown by scientists and the public, opened up a new field of study and launched the development of a new machine. The [Antiproton Decelerator \(AD\)](#) supplied several experiments with antiprotons from 2000 onwards. Just two years later, exactly 100 years after the birth of Paul Dirac, two collaborations [announced](#) the successful production of many “cold” antihydrogen atoms. Since then, experiments on antimatter have been increasingly successful, measuring the fundamental parameters of antiprotons and antihydrogen with ever greater precision. A second decelerator, ELENA, which came into service in 2020, further slows down the antiprotons to make it easier to study them.



Part of the detector used by the PS210 experiment, the first to produce antihydrogen atoms. (Image: [CERN](#))

## Recollections

Walter Oelert, then a researcher at the Jülich Research Centre in Germany, led the team who worked on the PS210 experiment. This experiment was rapidly assembled in 1995 to take advantage of the last weeks of the LEAR machine, in order to produce the world's first atoms of antihydrogen. He then took part in one of the pioneering experiments on antimatter in the early 2000s and was one of the initiators of the new ELENA antimatter decelerator.

"I'd like to quote Werner Heisenberg from 1972: "the discovery of antimatter was perhaps the

biggest jump of all the big jumps in physics in our century". He refers to the trendsetting predictions of Dirac, followed by the discovery and observation of the positron and antiproton in 1932 and 1955 respectively, with antihydrogen still awaiting its first observation.

During the early 1990s, the JETSET collaboration was running an experiment with antiprotons at LEAR. At a coffee break in autumn 1993, three accelerator scientists, Michel Chanel, Pierre Lefèvre and Dieter Möhl, reported that they had heard about an idea of how antihydrogen could be produced with just the experimental set-up that



the scientists were using on JETSET, with some additional detection systems. The JETSET collaboration was not interested in joining this enterprise, but after a few parasitic tests some of us were convinced that it must work and finally the LEAR programme committee allocated 48 hours of beam time for the experiment PS210 for producing antihydrogen atoms in flight.

No one in that small new collaboration will ever forget those exciting days in 1995. Every part of the experimental set-up would have to work; there would be no second chance. To add to the pressure, the press had already got hold of what was planned before we had any interactions that could possibly create the first antihydrogen atoms observed on Earth.



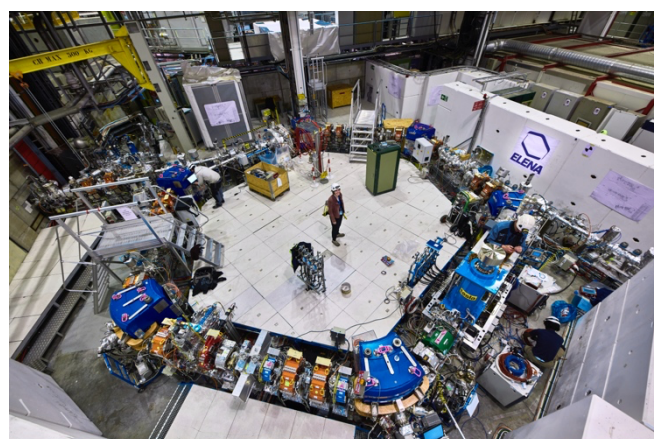
Walter Oelert, photographed here in 1996, led the team who created the first atoms of antihydrogen. (Image: [CERN](#))

We finally took data during a period of four weeks, two hours a day. Then it became really hectic. Offers from journals to publish the successful production of antihydrogen atoms were made even before the final results were available! A press release was prepared although the referee of the scientific paper was not willing to accept proof of the observation of antihydrogen atoms based only on the coincident registration of the annihilation signals from a positron and an antiproton, the constituents of an antihydrogen atom. However, further signals of energy and momentum determined during the experiment convinced both the referee and the editor, and the first observation of the production of atoms formed from antimatter particles could be announced. The public interest was enormous, the

result made it to headlines in many newspapers including the “New York Times” and “Der Spiegel”. One of the many interviews should be quoted:

- Journalist: Is it correct that the energy density of matter and antimatter is the highest one can think of?
- Oelert: Yes, certainly in principle that is true.
- Journalist: Does that mean that one can make an antimatter-bomb much much stronger than an atomic bomb?
- Oelert: Oh, such an application never ever came to my mind, let me think about it more precisely. (Short pause) No it is impossible for two reasons. It would be technically impossible to store such an amount of antimatter and the production efficiency would be much too low for any relevant amount of antimatter such that it can be used as an energy source. No way.

In the final article, the journalist wrote something like: “Oelert thinks about an antimatter-bomb”. But that is totally unrealistic, crazy science fiction and by no means my business. For basic science however, it was a great step towards the real work of producing ample amounts of cold antihydrogen and testing of fundamental symmetries. Seven years after this first observation, two experiments at the new [AD](#) complex successfully produced large numbers of cold antihydrogen atoms for the first time ever.



The ELENA decelerator, photographed in 2018. (Image: [CERN](#))

Approval for the ELENA upgrade was challenging. Luckily, after my presentation to the CERN Scientific Council, the Director-General rested a hand on my shoulder and said: “This was exactly the talk I wanted, thank you”. ELENA began in 2020 and now decelerates antiprotons from the

AD even further, allowing many more to be trapped by the experiments and ushering a new, exciting era of antimatter research. And now, 30 years after the fabrication of the first antihydrogen atoms, comparing hydrogen to antihydrogen atoms constitutes one of the best ways to make precise tests of differences between matter and antimatter. Their spectra are predicted to be

identical, so any tiny differences would open a window to new physics.”

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*This interview is adapted from the 2004 book “Infinitely CERN”, published to celebrate CERN’s 50th anniversary, updated with the help of Walter Oelert in 2024. Heisenberg’s quote appears in The Physicist’s Concept of Nature (1973), Vol. 1972, 271.*

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## CERN welcomes Estonia as its 24th Member State

Estonia becomes the first Baltic country to join CERN as a full Member State



*As Estonia becomes a full Member State of CERN, the Estonian flag is photographed along with the CERN flag (image: CERN)*

Geneva, 30 August 2024. Today, CERN is welcoming Estonia as its 24th Member State, marking the end of the formal application process that started in 2018 and crowning a period of cooperation that stretches back three decades.

“Estonia is delighted to join CERN as a full Member because CERN accelerates more than tiny particles, it also accelerates international scientific collaboration and our economies. We have seen this potential during our time as Associate Member State and we are keen to begin our full contribution,” said Alar Karis, President of Estonia. “On behalf of the CERN Council, I warmly welcome Estonia as the newest Member State of CERN,” said Eliezer Rabinovici, President of the CERN Council. “I am happy to see the community of CERN Member States enlarging, and I am looking forward to the enhanced participation of Estonia in the CERN Council and to its additional scientific contributions to CERN.”

“Estonia and CERN have been collaborating closely for some 30 years, and I am very pleased to

welcome Estonia to the ever-growing group of CERN Member States,” said Fabiola Gianotti, CERN Director-General. “I am sure the country and its scientific community will benefit from increased opportunities in fundamental research, technology development, and education and training.”

The bilateral relationship formally began in 1996, when Estonia and CERN signed a first cooperation agreement. A second such agreement, which further developed their scientific and technical cooperation, was concluded between the parties in 2010. On [19 June 2020](#), the parties signed an agreement concerning the granting to Estonia of the status of Associate Member State in the pre-stage to Membership of CERN, which entered into force on [1 February 2021](#).

Estonia has a broad engagement in the CERN scientific programme, and has been part of the CMS collaboration since 1997. Estonia’s [CMS](#) team participates in data analysis and the [Worldwide LHC Computing Grid](#) (WLCG), for which Estonia operates one of the Tier 2 centres, located in Tallinn. Scientists from Estonia also contribute to other experiments, including [CLOUD](#), [COMPASS](#), [NA66](#) and [TOTEM](#), and to studies for future colliders ([CLIC](#) and the Future Circular Collider, [FCC](#)). Estonian theorists are also very much involved in collaborations with CERN.

As a CERN Member State, Estonia will have voting rights in the Council, CERN’s highest decision-making authority. Membership will enhance opportunities for Estonian nationals to be recruited by CERN and for Estonian industry to bid for CERN contracts.



Science is CERN's primary mission, and Estonia's membership shows that the Organization's

exciting scientific programme continues to attract interest and support across the world.

## Refuse, reduce, reuse, recycle, repeat

**Discover CERN's dedicated storage, recuperation and sales facility, where you can buy second-hand furniture, IT equipment, metals, wood and more for your private or professional use**



*CERN's Storage, Recuperation and Sales service (Building 133), shown here in the foreground, can help us all to reduce our contribution to CERN waste. (Image: CERN)*

Waste is unquestionably a major environmental challenge worldwide and one we all play a role in tackling. Effective waste management is a fundamental pillar of CERN's efforts to minimise its environmental footprint.

Did you know that conventional non-hazardous waste represents 70% of CERN's total waste? Recycling is central to CERN's waste reduction strategy and the Laboratory remains committed to continuously improving its recycling rate: for reference, 69% of conventional waste was recycled in 2022 compared with 56% in 2018.

We all contribute to minimising waste, whatever the type. The [Storage, Recuperation and Sales Service \(Building 133\)](#) of the SCE department plays a key role in the refuse, reduce, reuse, recycle, repeat strategy. An unexpected hive of activity, this is the place where everything is prepared for reuse or recycling and sorted into the right channels. The service also resells or donates equipment that CERN no longer needs.

### **Don't buy new, reuse**

The service is a sales centre where you can buy second-hand furniture, IT equipment, metals, wood for DIY and more for your private use, or

collect for free for your professional CERN use. Browse the [online catalogue](#) and discover the many items available now. You can collect items at Building 133, by appointment only, every Thursday from 8.30 to 11.30 a.m. and from 1.30 to 4 p.m.

### **A place for everything and everything in its place**

Most of the hardware handled by the service comes from facilities that are no longer in operation, such as experiments or accelerators. The service also takes care of low value items from around the Laboratory and its offices, such as computers, furniture, various metals, all types of batteries, electronic and IT equipment, light sources (neon lights, bulbs), refrigeration appliances, paper and cardboard, plastic, glass, rubble, bulky items and wood.

The service is open weekdays from 8.30 to 11.30 a.m. and from 1.30 to 4 p.m. (closed Friday afternoons) and is available to anyone working on the CERN site. You can come and deposit your items directly or, if they are bulky, you can make a [transport request in EDH](#). For large volumes, the service can collect them using a grapple truck. If you have old furniture to dispose of, don't throw it in a skip: it can also be dismantled and recycled. Requests for furniture removal must be made using a transport request via EDH (category "Internal transport", type "Removal") and addressed to Building 133.

If you have any doubt about what waste goes where, check out this informative [webpage](#) or contact [recovery.sales@cern.ch](mailto:recovery.sales@cern.ch), and remember that there is a dedicated facility for handling [chemical waste](#) at Building 262.

Take a tour of the Building 133 [Storage, Recuperation and Sales service](#) facility in the video [here](#) or come and see it for yourself. And remember: the best waste is no waste.

*SCE department and HSE unit*

# CMS welcomes its new management

The new spokesperson, Gautier Hamel de Monchenault, and the two deputy spokespersons, Hafeez Hoorani and Anadi Canepa, will be representing CMS over the next two years



The new CMS spokesperson team: spokesperson Gautier Hamel de Monchenault (centre) and deputies Anadi Canepa (left) and Hafeez Hoorani (right). (Image: CMS collaboration)

The [CMS collaboration](#) welcomes its tenth management team: spokesperson Gautier Hamel de Monchenault and deputy spokespersons Anadi Canepa and Hafeez Hoorani. The new team began their mandate on 1 September 2024 and will represent the CMS collaboration until 31 August 2026. This is the tenth spokesperson team of the

CMS experiment since the signing of its letter of intent just over 30 years ago.

The next two years will represent a unique time in the history of CMS with the confluence of detector operations, analysis of unprecedented proton–proton and heavy-ion collision datasets, and the ambitious upgrade of the detector in preparation for the [High Luminosity phase of the LHC](#). Thanks to an exceptionally versatile particle physics detector and the ingenuity and creativity of a diverse collaboration, CMS will continue to address the most fundamental particle physics questions, including advancing research into the Higgs boson.

The CMS collaboration thanks the outgoing spokesperson Patricia McBride, and deputy spokespersons Lucia Silvestris and Wolfgang Adam for their incredible work from 2022 to 2024.

For detailed biographies of the new spokesperson team, visit the [CMS website](#).

*CMS collaboration*

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## Learn how to recognise and appreciate your efforts

Part six of the Work Well Feel Well series looks at the celebrating our achievements

### Building ourselves up brick by brick



With endless to-do lists, it can be easy to constantly focus on what still needs to be done. We forget that the efforts we have made, the paths we have taken and the experience we have acquired are just as important as the end result. At school, we were taught to expect congratulations based on exam success, rather than on the time invested. This means that, as

adults, if we don't achieve the expected result or if no one congratulates us, we can feel a sense of injustice or a total lack of recognition, which seriously demotivates us.

In the world of work, performance evaluations are still mainly based on whether an objective has been achieved. Employees may find themselves having to cover the responsibilities of an absent colleague, deal with a difficult individual, adapt to a change in procedure or abandon one project to focus on another, yet such extra efforts are all too often taken for granted.

Few of us have learned to acknowledge and value our own efforts, regardless of the result obtained. This lack of validation creates a state of permanent inner dissatisfaction, so we look to others for thanks or congratulations. External recognition

can encourage us, but the challenge is to develop inner self-recognition.

### Take action

As part of the “Efficiency and caring at work” campaign, the [Work Well Feel Well website](#) now offers **useful resources** that can be downloaded, including an [exercise](#) on **appreciating our daily efforts**.

Other suggestions include:

- **Creating “done” lists:** By regularly noting down what we have done, we not only realise how much we have accomplished but also learn to value our achievements and feel proud of ourselves. When a task has been completed, we can take a moment to appreciate our efforts.

- **Building new habits:** Throughout our careers, we will need to reflect on and value our achievements, whether we are applying for a job, preparing for a yearly appraisal or seeking a promotion. Getting into the habit of celebrating our own successes will help us to prepare for these career milestones.
- **Taking training courses:** The [CERN training catalogue](#) includes courses on [performance management](#) and [giving and receiving feedback](#) that can help us to self-evaluate and appreciate our own efforts.

*HR department*

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## How CERN IT keeps up with the data deluge

At CERN we are used to dealing with a deluge of data, but the numerical value behind the word “deluge” has significantly evolved over the years



*The new Prévessin Data Centre at CERN. (Image: CERN)*

At CERN, we are used to dealing with a deluge of data, but the numerical value behind the word “deluge” has significantly evolved over the years. Today, the average amount of collision data recorded on disk by the LHC experiments is a little under 3 petabytes (PB) per day, almost equal to the amount that was recorded in one month during Run 1. So far, the current 2024 proton run has written about 200 PB on disk. This figure is not far from the 204 PB of data recorded during the whole of Run 2 (2015–2018). Data is set to crescendo in future runs and with future

accelerators; the trend is already very clear but the details will evolve over the years.

In 2020, it became clear that with the Meyrin Data Centre alone, CERN would not be able to cope with all the data produced by the LHC experiments. The only viable solution to increase storage and processing capacity was to build a new data centre: the Prévessin Data Centre, [inaugurated](#) at the beginning of this year.

Fundamental differences exist between the two infrastructures: “The Meyrin Data Centre is highly robust and is fully covered by uninterruptible power supply (UPS) systems, but it can only support low power density in the racks,” explains Wayne Salter, Head of the Fabric group in the IT department, in charge of the procurement, management and operation of the data centres. “This means that it is better suited for low density equipment that needs guaranteed power, such as storage. In the new Prévessin Data Centre, where we don’t have full UPS coverage, we are concentrating the processing capacity – less vulnerable to power cuts – which is entirely made available to the [Worldwide LHC Computing Grid](#) (WLCG).”



All data produced at CERN still passes through the Meyrin Data Centre. It is the only place connected to all the experimental sites via the ultra-fast network of optical fibres. The data collection, copy and dispatching still happen here.

But beyond the data centres, the general upgrade of the IT services at CERN is a vital ongoing process. As part of this process, the Meyrin Data Centre Console service was [retired](#) at the end of March. Since then, both data centres have been monitored remotely with critical functions being handled by the CERN Control Centre (CCC) operators.



*While the Console service, shown here, was retired in March, the former operator room is gaining popularity as a filming location, with the Meyrin Data Centre as a backdrop. (Image: CERN)*

Hands-on, round-the-clock interventions are no longer required in the data centres thanks to a

combination of resilient, and in some cases, auto-remediating, IT services. “From the 1960s to recently, console operators held a critical role in data storage, processing and the operation of computing equipment. They managed expansive tape libraries, a custodian of data storage,” explains Olof Barring, Head of the Fabric group’s Infrastructure and Operations section, responsible for the operation of the data centres. “The manual operations and highly specialised skills of the console operators have underpinned the success of computing at CERN for decades. Their contribution has been invaluable and will go down in CERN history.”

What other revelations does the future hold? How will CERN cope with the growing data production rate of future experiments? “It is probably too early to predict the exact volume of data that future accelerators will produce and how we will deal with it, but we know that big challenges lie ahead and only careful, early planning will allow our scientific community to keep up with them”, replies Salter. To meet the needs of [High-Luminosity LHC](#), the IT department already foresees an expansion of the Prévessin Data Centre in 2027, making more space and power available, as well as the continued monitoring of the growing needs for the Meyrin Data Centre.

*Antonella Del Rosso*

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## Computer security: security workouts

Summertime. The moment to get in shape. Jogging. Step aerobics. CrossFit. Push-ups. Shadow-boxing. Zumba. Pilates. Whatever. Or to just stay as you are – fit and healthy – that would do too! And as summer comes to an end, here some suggestions for mental workouts. Cerebral challenges. Massaging your mind.

Just like your summer exercise regime, computer security requires [permanent training](#) to stay fit and healthy. Like for a marathon, it is not enough to know how to walk or run; training and exercising are a must. The body and mind need to be in sync and fit enough to take up the challenge. Again and again. Below you will find three

(hundred) suggestions for refreshing and improving your skills in computer security. Whether you are a computer engineer, physicist, engineer, lawyer, or work in HR, communication, etc.

### **The Zebra Alliance for everyone**

The Zebra Alliance hands-on training is a computer security incident simulation bringing together many different stakeholders of an organisation (here, the Zebra Alliance) to jointly respond to a critical computer security problem, including webmasters and security experts as well as PR, lawyers and HR<sup>[\[1\]](#)</sup>. Only through teamwork can the

origin, the culprit, be identified and contained. The last Zebra Alliance was run at the [end of May](#) and was so successful that we decided to run another one on 13 September. No IT knowledge beyond standard computer use is required. [Some places are still available.](#)

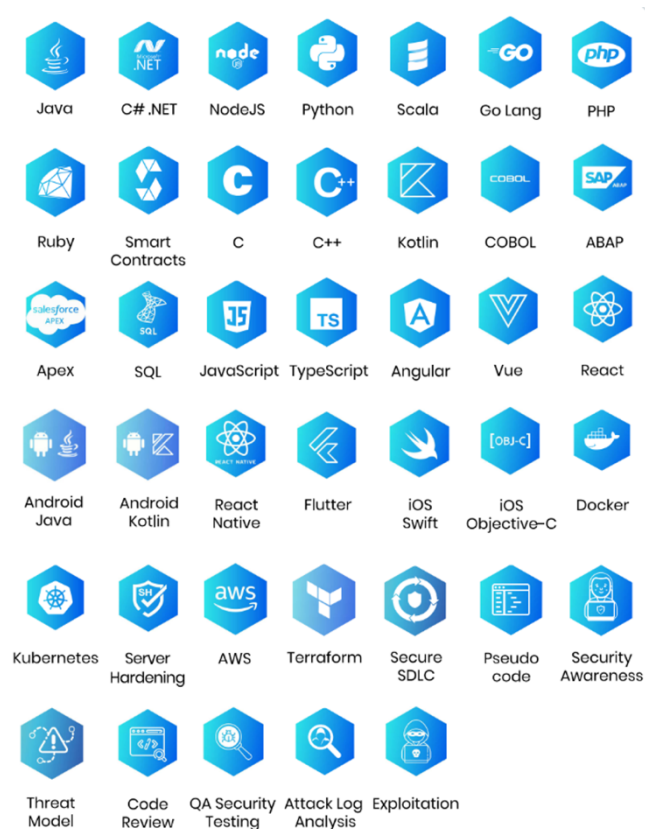
### The WhiteHat Challenge for newcomers in penetration testing

If you want to become a real “hacker”, learning vulnerability scanning and penetration testing, and dig deep into computer security practices, attend the WhiteHat Training in October. Only a laptop, some basic knowledge of web technologies and a curious mind are required. [Session #1 on 2 October](#) will give you an introduction to ethical hacking and the ground rules of the WhiteHat Challenge before delving into the use and abuse cases of web applications. Afterwards, trained with the right tools, knowledge and strategies, you can pit yourself against our inherently very vulnerable Movies website. This capture-the-flag challenge has many levels, starting with the easy ones and then allowing you to dig deeper and deeper – until either you crack the website completely or your level of expertise is reached. [Session #2](#) is held two weeks later, on 16 October, designed to resolve the riddles of this capture-the-flag so you can verify your findings or marvel at how to break into a web server. Are you ready to become a penetration tester?

### SecureFlag’s hands-on courses for software developers, programmers and IT service managers

For software developers, programmers, web administrators, IT service managers and anyone else who regularly touches code, we strongly recommend subscribing to the SecureFlag training platform. It provides hands-on courses, exercises and virtual environments for you to improve your skills in secure software development in any of your favorite programming language(s); to securely configure your systems, virtual machines and containers and securely operate your web and computing services (watch the [demo video](#)). The licence scheme is “all-you-can-eat”, meaning that

with a valid licence you can access all 30 labs (see image below) and hundreds of security-dedicated courses. Licences can be requested via [CERN's Learning Hub](#) and are valid until 31 May 2025 – so the earlier you sign up, the more food for your brain.



Thank you very much for participating in our training sessions, for understanding what to do in the event of a security incident, for improving your code, your application, your IT service. And for helping to secure the Organization!

[\[1\]](#) In a previous training session we even had participants from the Geneva police and the Pays-de-Gex gendarmerie.

*Computer Security team*

*Do you want to learn more about computer security incidents and issues at CERN? Follow our [Monthly Report](#). For further information, questions or help, check [our website](#) or contact us at [Computer.Security@cern.ch](mailto:Computer.Security@cern.ch).*

# Announcements

## Upcoming events

Check out this curated list of upcoming events for the CERN community

**10 Sep 16:30** | Knowledge Sharing | Online | CERN Alumni Network | EN

[From research to industry - selling your skills to a future employer \(joint event with PSI\)](#)

**10 Sep 17:30** | Knowledge Sharing | Council Chamber | Staff Association | FR

[« Perdre le Nord », par Bernard Voyer](#)

**17 Sep 17:30** | At CERN | R1 lawn | CERN70 | EN & FR  
[CERN70 Community Event](#) (more [CERN70](#) events)

**18 Sep 10:00** | Knowledge Sharing | Online | CERN Procurement | EN

[CERN Industry Webinar](#)

**19 Sep 11:00** | Knowledge Sharing | CERN Library | Archives, Library and Open Science events | EN

[Meet the authors of "The Future of the Large Hadron Collider"](#)

**24 Sep 11:00** | Knowledge Sharing | CERN Library | Archives, Library and Open Science events | EN

[The Voice of Science – Meet Dr. Katie Mack \(AstroKatie\)](#)

**1 Oct 15:00** | Knowledge Sharing | Online | CERN70 | EN

[CERN70 official ceremony: Inspiring the future](#) (more [CERN70](#) events)

*We invite our CERN community to follow the event online. Attendance by invitation only.*

**4 Oct 18:00** | Knowledge Sharing | CERN Science Gateway | Public events | FR

[Partage ta science – 2024](#)

**11 Oct 14:00** | Knowledge Sharing | Online | HR Learning and Development | EN

[L&D Micro-talk 12 – Demystifying creativity](#)

**17 Oct 13:30** | Knowledge Sharing | CERN Library | Archives, Library and Open Science events | EN

[Meet the author of "Relic Gravitons"](#)

**18 Oct 14:00** | Knowledge Sharing | Main Auditorium | Commemorations | EN

[Symposium to celebrate Carlo Rubbia's 90th birthday](#)

**Registration now open**

**22 Sep–5 Oct** | Accelerators | Santa Susanna (ES) | CERN Accelerator School | EN

[Introduction to Accelerator Physics](#)

**27 Nov–6 Dec** | Experiments | CERN | DRD1 | EN

[DRD1 Gaseous Detectors School](#)

This is a curated list of events relevant to the CERN community. More events are available here: [home.cern/events](https://home.cern/events)

Indico also shows ALL events happening [today, this week](#) and in a [calendar view](#).

If you would like your event to appear on an upcoming Bulletin events list, please contact [bulletin-editors@cern.ch](mailto:bulletin-editors@cern.ch)

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## Entrance C reopens from 9 September

From Monday 9 September 2024, Entrance C will be operational again and will be open during normal opening hours, Monday to Friday from 7 a.m. to 7 p.m.

In order to install the new gate, Entrance C will be closed to vehicles, bicycles and pedestrians from Monday 2 September at 2 p.m. to Wednesday 4 September at 2 p.m.

Entrance C has experienced disruptions since spring, due to [car accident on 10 April](#), and a [delay in receiving replacement parts](#).

*SCE department*



# REMINDER: Discover CERN70 with the upgraded CERN Campus app

We are thrilled to announce significant updates to the CERN Campus app, just in time for our upcoming [CERN70 Community Event](#) on 17 September.

Launched a year ago, the app has already surpassed 10 000 downloads and attracts an average of 600 daily users.

The app offers a comprehensive suite of features including emergency notifications for safety updates, direct access to CERN's emergency services via an SOS button, an interactive map and the campus phonebook. Users can also check restaurant menus and occupancy level, as well as the operating hours of different campus services; stay informed about news and events; and access campus services such as housing, mobility, CERN

clubs, CERN market, the CAGI cultural kiosk, library resources and more.

Recently updated to include all essential details for the upcoming CERN70 Community Event on 17 September, the app continues to enhance user experience and accessibility, offering a complete overview of the programme, schedule, artists, installations in the "Village", maps and practical information. The app is now accessible without a CERN login (except for the phonebook), making it easier for newcomers, contractors and professional visitors to engage in activities at CERN.

Explore the new features of the CERN Campus app by [downloading it here](#).

*SCE department*

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## Obituaries

### Max Klein (1951–2024)



(Image: [University of Liverpool](#))

Born in Berlin, Max earned his Diploma in Physics in 1973 from Humboldt University of Berlin (HUB, East Germany, GDR), with a thesis on low-energy heavy-ion physics. He received his PhD in 1977

from the Institute for High Energy Physics (IHEP) of the Academy of Sciences of the GDR in Zeuthen (now part of DESY) on the subject of multiparticle production, and his "habilitation" degree in 1984 from HUB.

From 1973 to 1991, he conducted research at IHEP Zeuthen, spending several years from 1977 at the Joint Institute for Nuclear Research (JINR) in Dubna, and from the 1980s at DESY and CERN. He was awarded the Max von Laue Medal by the Academy of Sciences of the GDR in 1985 for his role in determining the asymmetric interaction of polarised positive and negative muons with the NA4 muon spectrometer at CERN's SPS M2 muon beam.

Max worked at DESY from 1992 to 2006. As a member of the H1 experiment at the lepton-proton collider HERA from 1985, his research focused on investigating the internal structure of protons using deep inelastic scattering, and he served as H1 spokesperson from 2002 to 2006.

He became a Professor in the Department of Physics at the University of Liverpool in 2006 and joined the ATLAS collaboration in 2007. Within ATLAS, he made key contributions to data analysis, notably on the high-precision 7 TeV inclusive W and Z boson production cross sections and associated properties. He led the Liverpool ATLAS team for eight years, from 2009 to 2017. Under his guidance, the 30-strong group contributed to the maintenance of the SCT detector, as well as to ATLAS data preparation, precision SM measurements, Higgs physics studies, new physics searches and more. His contributions to understanding the proton structure using deep-inelastic scattering led to the award of the Max Born Prize by the Deutsche Physikalische Gesellschaft (DPG) and the Institute of Physics (IOP) in 2013.

Max had a unique ability to form collaborations, bringing together people from different backgrounds to work towards a common goal. From 2008 to 2022, he led the LHeC study, an electron beam upgrade of the LHC. Following the Higgs boson discovery in 2012, the CERN Management gave him the official mandate to develop this study and included LHeC in the European Strategy of Particle Physics discussions and as an integral part of the FCC study. Max was also a strong advocate for developing energy

recovery linear (ERL) accelerators and was influential in the development of the PERLE ERL demonstrator accelerator at IJCLab, for which he acted as spokesperson up to 2023.

Besides being a distinguished scientist, he was a man of unwavering principles, grounded in his selfless interactions with others and his deep sense of humanity. His visions were rooted in realism. Drawing from his experience as a bridge between East and West, he was a strong advocate for international scientific collaboration and the responsibility of scientists toward their societies.

The many of us who were fortunate enough to have worked alongside Max over the years know how profound his knowledge of physics was and how dedicated he was to experimental particle physics. We admired his ability to mentor and support students, postdocs and early-career researchers, as well as his wise and calm approach to solving problems. Throughout his long and exceptional career, Max worked with passion on theory, detectors, accelerators and data analysis. Max's passing is a profound loss for ATLAS and the entire high-energy physics community, but his legacy will endure. Our deepest condolences go to his family, especially to his beloved wife and our close colleague, Uta.

*His friends and colleagues*

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## Ombud's corner

### Three life skills are essential at CERN: politeness, respect and gratitude

"He doesn't even say hello to me anymore." "She doesn't listen to my opinion." "Whatever I do, it's not enough."

I often hear these kinds of emotionally charged complaints, with people feeling that their very identity is at stake. Hence the idea for this article on three life skills that enable individuals to interact effectively and harmoniously with others.

- Politeness, first and foremost, is vital both when being introduced to strangers and in our daily interactions with colleagues.

Politeness may seem outdated in this era of instant communication. But when it is lacking, when simple acts like smiling and saying "hello", "thank you" and "bye" fall by the wayside, we feel its absence. Politeness plays a crucial role in interpersonal communication and relations, helping to create a climate of trust and respect.

- Respect "changes your life", as we were reminded by the slogan of the Geneva-based campaign that was picked up by [CERN in 2010](#). When two people respect each other, they:

- are polite to each other;
- accept that they are different but interact as equals;
- speak to each other openly and honestly.

Respect is essential in all human interactions, whether personal or professional. To that end, also in 2010, CERN established a [Code of Conduct](#) – a practical guide to help us understand how to behave, how to treat others and how to expect to be treated, in accordance with CERN values.



### *It's all about respect*

- Gratitude, lastly, is an act of recognition and appreciation towards others for considerate behaviour, support or help. It may also extend to

appreciation for experiences, happy moments or things happening in our lives.

Our social life has a major impact on our well-being and our mental health. Being polite, showing respect, demanding respect where necessary and thanking those who stand by us are all ways to improve our daily lives by positively influencing our relationships in the workplace and beyond.

If you would like to talk more about this subject or its implications in a particular situation, please come and see me. As Ombud, my main purpose is to provide a non-official channel for conflict resolution, as well as to offer my opinion on the implementation of the Code of Conduct.

Thank you for your attention, and I look forward to seeing you soon.

*Marie-Luce Falipou*

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*I would like to hear your reactions and suggestions – join the CERN Ombud Mattermost team at <https://mattermost.web.cern.ch/cern-ombud/>. The Ombud is available from Monday to Friday in office B500/1-004 on the Meyrin site. To make an appointment, in person or online, contact the Ombud at [ombuds@cern.ch](mailto:ombuds@cern.ch).*

*More information can be found on the Ombud's website: <https://ombuds.web.cern.ch>*